

PHOTOGRAPHY SAAS & DIGITAL ASSET MANAGEMENT (DAM)

REST API SPECIFICATION & REFERENCE MANUAL | VERSION 2.0

Base URL: `http://<domain-or-ip>:8000/api/` • **Auth:** Bearer JWT • **Async Engine:** Celery 5.6.3 + Redis • **Framework:** Django 6.0.7 / DRF 3.17.1

TABLE OF CONTENTS

- 1. EXECUTIVE API ARCHITECTURE & CONVENTIONS
- 2. AUTHENTICATION & IDENTITY SUBSYSTEM (/api/auth/)
- 3. PHOTOGRAPHER STUDIO, BRANDING & NOTIFICATIONS (/api/photographers/)
- 4. SUBSCRIPTIONS, TIERS & BILLING ENGINE (/api/subscriptions/)
- 5. ENTERPRISE CLIENT GALLERIES & ASSET MANAGEMENT (/api/storage/galleries/)
- 6. DIRECT-TO-CLOUD STORAGE & UPLOAD PIPELINE
- 7. CLIENT PUBLIC ACCESS, PIN SECURITY & PROOFING WORKFLOW
- 8. DOWNLOAD PIPELINE & ASYNCHRONOUS BULK ZIP
- 9. AI BIOMETRIC FACE RECOGNITION & DISCOVERY
- 10. STORAGE ACCOUNT METRICS & QUOTA AUDIT
- 11. EVENT PHOTO SHARING & GUEST QR ACCESS (/api/storage/events/ & /api/storage/share/)
- 12. RATE LIMITING, SECURITY PROTOCOLS & ERROR CODES

1. EXECUTIVE API ARCHITECTURE & CONVENTIONS

1.1 URL Conventions & Aliasing:

- Master Base URL: ``/api/``
- Subsystems are organized under distinct modular prefixes:
- ``/api/auth/`` -> User identity, multi-identifier login, JWT, OTP
- ``/api/photographers/`` -> Studio profile, watermark branding, feed, notifications
- ``/api/subscriptions/`` -> Tiered plans, quotas, billing upgrades
- ``/api/storage/`` -> Enterprise galleries, direct uploads, proofing, downloads
- Backward-compatibility aliases are preserved at the root level:
- ``/api/galleries/`` -> maps directly to ``/api/storage/galleries/``
- ``/api/events/`` -> maps directly to ``/api/storage/events/``
- ``/api/share/`` -> maps directly to ``/api/storage/share/``
- ``/api/account/storage/`` -> maps directly to ``/api/storage/account/storage/``

1.2 Request & Response Formats:

- All JSON payloads must be sent with ``Content-Type: application/json``.
- Multipart payloads (avatars, watermark stamps, manual uploads, selfies) must use ``Content-Type: multipart/form-data``.
- Datetimes are formatted in ISO-8601 UTC (e.g., ``2026-09-12T12:00:00Z``).
- IDs: User & Photographer IDs are Integers; Galleries, Media, Selections, Reservations, and Bulk Jobs use UUID v4 strings.

1.3 Authentication Header:

- Format: ``Authorization: Bearer <access_token>``
- Access tokens have a 60-minute lifespan; refresh tokens have a 1-day lifespan.

2. AUTHENTICATION & IDENTITY SUBSYSTEM (/api/auth/)

2.1 Check Username Availability

Method: **POST**

Endpoint: /api/auth/check-username/

Auth: None

Request Body (JSON):

```
{
  "username": "lumiere_studio"
}
```

Response (200 OK):

```
{
  "status": "success",
  "available": true,
  "message": "Username is available"
}
```

2.2 Check Identifier (Email or Phone)

Method: **POST**

Endpoint: /api/auth/check-identifier/

Auth: None

Request Body (JSON):

```
{
  "identifier": "studio@lumieremedia.com"
}
```

Response (200 OK):

```
{
  "status": "success",
  "available": true,
  "message": "Identifier is available"
}
```

2.3 Register New User / Studio Account

Method: **POST**

Endpoint: /api/auth/register/

Auth: None

Request Body (JSON):

```
{
  "username": "sarang_varma",
  "email": "sarang@atelierstudio.com",
  "phone": "+3902998877",
  "password": "SecureStudioPassword123!",
  "fullname": "Sarang Varma",
  "role": "photographer" // Choices: "photographer", "user", "staff", "admin"
}
```

Response (201 Created):

```
{
  "status": "success",
  "message": "Registration successful. OTP sent.",
  "user_id": 12
}
```

2.4 Verify Registration OTP

Method: **POST**

Endpoint: /api/auth/verify-otp/

Auth: None

Request Body (JSON):

```
{
  "email": "sarang@atelierstudio.com",
  "otp": "839201"
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Account verified successfully."
}
```

2.5 Resend Registration OTP

Method: POST**Endpoint:** /api/auth/resend-otp/**Auth:** None**Request Body (JSON):**

```
{
  "email": "sarang@atelierstudio.com"
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "OTP resent successfully."
}
```

2.6 User Login (Multi-Identifier: Username, Email, or Phone)

Method: POST**Endpoint:** /api/auth/login/**Auth:** None**Request Body (JSON):**

```
{
  "identifier": "sarang@atelierstudio.com", // Accepts username, email, or phone
  "password": "SecureStudioPassword123!"
}
```

Response (200 OK):

```
{
  "status": "success",
  "tokens": {
    "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9...",
    "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
  },
  "user": {
    "id": 12,
    "unique_id": "40a597a7-33e1-4c6e-b3f8-62d26f98c92a",
    "username": "sarang_varma",
    "email": "sarang@atelierstudio.com",
    "fullname": "Sarang Varma",
    "role": "photographer",
    "is_email_verified": true
  }
}
```

2.7 Refresh JWT Access Token

Method: POST**Endpoint:** /api/auth/token/refresh/**Auth:** None**Request Body (JSON):**

```
{
  "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Response (200 OK):

```
{
  "access": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

2.8 Logout (Blacklist Refresh Token)

Method: POST**Endpoint:** /api/auth/logout/**Auth:** Bearer Token**Request Body (JSON):**

```
{
  "refresh": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9..."
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Logged out successfully."
}
```

2.9 Check Login State

Method: GET**Endpoint:** /api/auth/check-login/**Auth:** Bearer Token**Response (200 OK):**

```
{
  "is_authenticated": true,
  "user_id": 12,
  "username": "sarang_varma",
  "role": "photographer"
}
```

2.10 Password Reset via OTP

Step 1 - Request OTP:

POST /api/auth/reset-password/otp/

Body: { "identifier": "sarang@atelierstudio.com" }

Response: { "status": "success", "message": "Password reset OTP sent." }**Step 2 - Resend OTP:**

POST /api/auth/reset-password/resend-otp/

Body: { "identifier": "sarang@atelierstudio.com" }

Step 3 - Verify OTP & Set New Password:

POST /api/auth/reset-password/verify-otp/

Body:

```
{
  "identifier": "sarang@atelierstudio.com",
  "otp": "492019",
  "new_password": "NewSecureStudioPassword456!"
}
```

Response: { "status": "success", "message": "Password updated successfully." }

2.11 Direct Password Reset (Development / Administrative)

Method: POST**Endpoint:** /api/auth/reset-password/direct/**Auth:** None

Body: { "identifier": "sarang@atelierstudio.com", "new_password": "DirectPass123!" }

2.12 Authenticated Password Change

Method: POST

Endpoint: `/api/auth/change-password/`

Auth: Bearer Token

Body (JSON):

```
{
  "old_password": "CurrentPassword123!",
  "new_password": "BrandNewPassword456!"
}
```

Response (200 OK):

```
{ "status": "success", "message": "Password changed successfully." }
```

2.13 Google OAuth2 Sign-In / Token Exchange

Method: **POST**

Endpoint: `/api/auth/google/`

Auth: None

Body (JSON):

```
{
  "token": "<google_id_token>"
}
```

Response (200 OK): Returns JWT access/refresh pair and user profile.

2.14 Passwordless Authentication: Send OTP (Enter Email)

Method: **POST**

Endpoint: `/api/auth/passwordless/send-otp/` (or alias `/api/auth/otp/send/`)

Auth: None

Request Body (JSON):

```
{
  "email": "sarang@atelierstudio.com"
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Verification code sent to your email for login.",
  "email": "sarang@atelierstudio.com",
  "is_registered": true // false if new user, true if existing user
}
```

2.15 Passwordless Authentication: Verify OTP (Login or Auto-Register)

Method: **POST**

Endpoint: `/api/auth/passwordless/verify-otp/` (or alias `/api/auth/otp/verify/`)

Auth: None

Request Body (JSON):

```
{
  "email": "sarang@atelierstudio.com",
  "otp": "839201",
  "fullname": "Sarang Varma", // Optional: Used if new user is created
  "role": "photographer" // Optional: "photographer" | "user" (default "photographer")
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Login successful.", // or "Account registered and logged in successfully"
  "is_new_user": false,          // true if account was auto-created, false if existing
  "tokens": {
    "access": "<jwt_access_token>",
    "refresh": "<jwt_refresh_token>"
  },
  "user": {
    "id": 12,
    "unique_id": "40a597a7-33e1-4c6e-b3f8-62d26f98c92a",
    "username": "sarang_varma",
    "email": "sarang@atelierstudio.com",
    "fullname": "Sarang Varma",
    "role": "photographer",
    "is_email_verified": true
  }
}
```

2.16 Passwordless Authentication: Resend OTP**Method:** POST**Endpoint:** /api/auth/passwordless/resend-otp/ (or alias /api/auth/otp/resend/)**Auth:** None**Request Body (JSON):**

```
{
  "email": "sarang@atelierstudio.com"
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "A new verification code has been sent to your email.",
  "email": "sarang@atelierstudio.com",
  "is_registered": true
}
```

3. PHOTOGRAPHER STUDIO, BRANDING & NOTIFICATIONS (/api/photographers/)**3.1 Get Current Photographer Profile & Branding****Method:** GET**Endpoint:** /api/photographers/profiles/me/**Auth:** Bearer Token (Photographer)

Response (200 OK):

```
{
  "id": 1,
  "studio_name": "LUMIÈRE STUDIO & ATELIER",
  "name": "Sarang Varma",
  "avatar": "http://api.domain.com/media/photographer_profiles/avatar.jpg",
  "bio": "Editorial wedding & haute couture photography based in Milan.",
  "phone": "+39 02 9988 7766",
  "email": "sarang@atelierstudio.com",
  "location": "Milan & Lake Como, Italy",
  "website_url": "https://lumierestudio.it",
  "instagram_handle": "lumiere.atelier",
  "default_template": "editorial",
  "enable_watermark": true,
  "watermark_text": "@ Lumière Studio",
  "watermark_image": null,
  "watermark_opacity": 0.45,
  "watermark_position": "bottom_right",
  "is_onboarded": true,
  "onboarding_step": 3,
  "storage_used_bytes": 12884901888,
  "storage_reserved_bytes": 0,
  "subscription": {
    "id": "e8d42398-b80c-43f1-bd12-fbf3e0a29481",
    "plan_tier": "pro",
    "plan_name": "Pro Studio Annual",
    "status": "active",
    "expires_at": "2027-09-12T00:00:00Z"
  }
}
```

3.2 Update Photographer Profile & Studio Details**Method:** PUT or PATCH**Endpoint:** /api/photographers/profiles/me/update/**Auth:** Bearer Token (Photographer)**Request Body (JSON):**

```
{
  "studio_name": "ATELIER & CO. FINE ART",
  "name": "Sarang Varma",
  "bio": "Fine art and destination weddings across Europe.",
  "location": "Paris & Milan",
  "website_url": "https://atelierfineart.com",
  "instagram_handle": "atelier.fineart",
  "default_template": "masonry"
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Profile updated successfully.",
  "data": { ... }
}
```

3.3 Upload / Update Studio Avatar**Method:** POST**Endpoint:** /api/photographers/profiles/me/avatar/**Auth:** Bearer Token (Photographer)

Content-Type: multipart/form-data

Body:

avatar: <binary image file (JPEG/PNG/WebP)>

Response (200 OK):

```
{
  "status": "success",
  "message": "Avatar updated successfully.",
  "avatar_url": "http://api.domain.com/media/photographer_profiles/avatar.jpg"
}
```

3.4 Configure Studio Watermark Suite

Method: PATCH**Endpoint:** /api/photographers/profiles/me/watermark/**Auth:** Bearer Token (Photographer)

Content-Type: application/json or multipart/form-data

Request Body:

```
{
  "enable_watermark": true,
  "watermark_text": "© Atelier Lumière Fine Art",
  "watermark_opacity": 0.40,
  "watermark_position": "bottom_right" // Choices: "center", "bottom_right", "bottom_left", "repeated"
  // watermark_image: <optional PNG file with transparency for logo stamp>
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Watermark settings updated successfully.",
  "data": {
    "enable_watermark": true,
    "watermark_text": "© Atelier Lumière Fine Art",
    "watermark_image": null,
    "watermark_opacity": 0.4,
    "watermark_position": "bottom_right"
  }
}
```

3.5 Set Up Your Profile & Onboarding (Screen: Step 3 of 3)

Method: GET or POST**Endpoint:** /api/photographers/onboarding/ (or /api/photographers/onboarding/complete/ or /api/photographers/onboarding/setup/)**Auth:** Bearer Token (Photographer)

Content-Type: application/json or multipart/form-data

GET Request (Inspect current onboarding status & profile details):

Response (200 OK):

```
{
  "status": "success",
  "is_onboarded": false,
  "onboarding_step": 1,
  "name": "Sarang Varma",
  "phone": "",
  "occupation": "",
  "avatar_url": ""
}
```

POST Request Body (JSON or multipart/form-data):

```
{
  "name": "Sarang Varma", // Your Name (required)
  "phone": "+1 (555) 000-0000", // Phone Number (optional)
  "occupation": "Wedding Photographer", // Occupation (e.g. Wedding Photographer, Fashion, etc.)
  "avatar": <optional custom image file upload>, // Profile Picture (PNG/JPG/WEBP up to 5MB, optional / not mandatory)
  "avatar_url": "https://.../avatar.jpg", // Or optional avatar image URL
  "onboarding_step": 3 // Step index (default: 3)
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Profile setup and onboarding completed successfully.",
  "is_onboarded": true,
  "onboarding_step": 3,
  "profile": {
    "id": 1,
    "name": "Sarang Varma",
    "phone": "+1 (555) 000-0000",
    "occupation": "Wedding Photographer",
    "avatar_url": "http://api.domain.com/media/photographer_profiles/custom_avatar.jpg"
  },
  "is_onboarded": true,
  "onboarding_step": 3
},
"user": {
  "id": 12,
  "username": "sarang_varma",
  "email": "sarang@atelierstudio.com",
  "fullname": "Sarang Varma",
  "phone": "+1 (555) 000-0000"
}
}
```

3.6 Studio Notifications List**Method:** GET**Endpoint:** /api/photographers/notifications/ (or /api/photographers/notifications/list/)**Auth:** Bearer Token (Photographer)**Response (200 OK):**

```
[
  {
    "id": 101,
    "title": "Client Proofing Submitted",
    "message": "Claire & Julian submitted 24 favorites for Lake Como Villa Balbianello",
    "event_type": "proofing_submitted",
    "related_gallery_id": "89b91e92-e7f0-4fa8-a1b7-d1e9f1938910",
    "is_read": false,
    "created_at": "2026-09-12T10:14:00Z"
  }
]
```

3.7 Mark Single Notification as Read**Method:** POST or PATCH**Endpoint:** /api/photographers/notifications/<int:id>/read/**Auth:** Bearer Token (Photographer)**Response (200 OK):**

```
{ "status": "success", "message": "Notification marked as read." }
```

3.8 Mark All Notifications as Read**Method:** POST**Endpoint:** /api/photographers/notifications/read-all/**Auth:** Bearer Token (Photographer)**Response (200 OK):**

```
{ "status": "success", "message": "All notifications marked as read." }
```

3.9 Notification Preferences

Get Preferences:

Method: GET**Endpoint:** /api/photographers/notifications/preferences/**Auth:** Bearer Token

Response (200 OK):

```
{
  "notify_client_visited": true,
  "notify_photos_downloaded": true,
  "notify_favorites_selected": true,
  "notify_storage_alerts": true,
  "notify_marketing_updates": false
}
```

Update Preferences:

Method: **PUT or PATCH**

Endpoint: **/api/photographers/notifications/preferences/**

Auth: Bearer Token

Body (JSON):

```
{
  "notify_client_visited": true,
  "notify_photos_downloaded": false
}
```

Response (200 OK):

```
{ "status": "success", "data": { ... } }
```

3.10 Portfolio Categories & Showcase Posts

- GET /api/photographers/categories/list/ (List categories)
- POST /api/photographers/categories/create/ (Create category)
- GET /api/photographers/posts/ (Public showcase feed)
- POST /api/photographers/posts/create/ (Create showcase post)
- GET /api/photographers/posts/me/ (Studio's own posts)
- GET /api/photographers/posts/<id>/ (Detail of showcase post)
- PUT /api/photographers/posts/<id>/update/ (Update showcase post)
- DELETE /api/photographers/posts/<id>/delete/ (Delete showcase post)
- GET /api/photographers/feedbacks/ (Client reviews)
- POST /api/photographers/feedbacks/create/ (Submit review)

4. SUBSCRIPTIONS, TIERS & BILLING ENGINE (/api/subscriptions/)

4.1 List Subscription Plans (Pricing Matrix)

Method: **GET**

Endpoint: **/api/subscriptions/plans/**

Auth: None

Response (200 OK):

```

{
  "count": 4,
  "results": [
    {
      "id": 1,
      "name": "Starter Studio",
      "tier": "starter",
      "billing_cycle": "monthly",
      "price_monthly": "19.00",
      "storage_limit_bytes": 53687091200,      // 50 GB
      "storage_limit_gb": 50,
      "max_galleries": 10,
      "allowed_templates": ["editorial", "masonry"],
      "video_delivery_enabled": false,
      "face_search_enabled": false,
      "watermark_customization": false,
      "features_list": [
        "50 GB High-Speed Storage",
        "Up to 10 Active Client Galleries",
        "Editorial & Masonry Layouts",
        "Standard Client Proofing"
      ]
    },
    {
      "id": 2,
      "name": "Pro Studio Annual",
      "tier": "pro",
      "billing_cycle": "annual",
      "price_monthly": "39.00",
      "storage_limit_bytes": 128849018880,      // 120 GB
      "storage_limit_gb": 120,
      "max_galleries": 0,                      // 0 = Unlimited
      "allowed_templates": ["editorial", "masonry", "cinematic", "minimal"],
      "video_delivery_enabled": true,
      "face_search_enabled": true,
      "watermark_customization": true,
      "features_list": [
        "120 GB Cloud Storage",
        "Unlimited Galleries",
        "All 4 Bespoke Layout Templates",
        "4K Video Asset Delivery",
        "AI Biometric Face Search",
        "Custom Watermark Branding Suite"
      ]
    },
    {
      "id": 3,
      "name": "Studio Enterprise",
      "tier": "studio",
      "billing_cycle": "annual",
      "price_monthly": "89.00",
      "storage_limit_bytes": 536870912000,      // 500 GB
      "storage_limit_gb": 500,
      "max_galleries": 0,
      "allowed_templates": ["editorial", "masonry", "cinematic", "minimal"],
      "video_delivery_enabled": true,
      "face_search_enabled": true,
      "watermark_customization": true,
      "features_list": [
        "500 GB Cloud Storage",
        "Dedicated CDN Edge",
        "Selective High-Speed Bulk ZIP Export",
        "Custom Domain Support"
      ]
    },
    {
      "id": 4,
      "name": "Master Atelier 1TB",
      "tier": "master",
      "billing_cycle": "annual",
      "price_monthly": "149.00",
      "storage_limit_bytes": 1099511627776,      // 1 TB
      "storage_limit_gb": 1024,
      "max_galleries": 0,
      "allowed_templates": ["editorial", "masonry", "cinematic", "minimal"],
      "video_delivery_enabled": true,
      "face_search_enabled": true,
      "watermark_customization": true,
    }
  ]
}

```

```

    "features_list": [
      "1 TB Ultra Storage",
      "VIP 24/7 Concierge Support",
      "Unlimited Sub-users & Assistants"
    ]
  }
}

```

4.2 Current Subscription & Quota Status

Method: **GET**

Endpoint: **/api/subscriptions/current/**

Auth: Bearer Token (Photographer)

Response (200 OK):

```

{
  "id": "e8d42398-b80c-43f1-bd12-fbf3e0a29481",
  "status": "active",
  "started_at": "2026-09-12T07:15:00Z",
  "expires_at": "2027-09-12T07:15:00Z",
  "auto_renew": true,
  "plan": {
    "id": 2,
    "name": "Pro Studio Annual",
    "tier": "pro",
    "billing_cycle": "annual",
    "price_monthly": "39.00",
    "storage_limit_bytes": 128849018880,
    "storage_limit_gb": 120,
    "allowed_templates": ["editorial", "masonry", "cinematic", "minimal"],
    "video_delivery_enabled": true,
    "face_search_enabled": true,
    "watermark_customization": true
  },
  "storage_used_bytes": 12884901888,
  "storage_reserved_bytes": 15000000,
  "storage_limit_bytes": 128849018880,
  "usage_percentage": 10.0
}

```

4.3 Initiate Plan Upgrade / Cycle Change

Method: **POST**

Endpoint: **/api/subscriptions/upgrade/**

Auth: Bearer Token (Photographer)

Request Body (JSON):

```

{
  "plan_id": 3,
  "billing_cycle": "annual",
  "payment_gateway_ref": "pay_stripe_cs_test_a1b2c3d4"
}

```

Response (200 OK):

```

{
  "status": "success",
  "message": "Subscription upgraded to Studio Enterprise.",
  "subscription": { ... }
}

```

5. ENTERPRISE CLIENT GALLERIES & ASSET MANAGEMENT (/api/storage/galleries/)

5.1 Create Client Gallery

Method: **POST**

Endpoint: **/api/storage/galleries/ (or /api/galleries/)**

Auth: Bearer Token (Photographer)

Request Body (JSON):

```
{
  "title": "Amalfi Coast Villa Cimbrone Editorial",
  "client_name": "Elena & Matteo Rossi",
  "client_email": "elena.rossi@wedding.it",
  "event_date": "2026-06-18",
  "description": "Fine art sunset ceremony overlooking the Tyrrhenian Sea.",
  "template_id": "editorial",          // "editorial" | "masonry" | "cinematic" | "minimal"
  "visibility": "public",              // "public" | "private" | "password_protected"
  "is_password_protected": true,
  "password": "Cimbrone2026!",        // Optional client PIN/password
  "allow_downloads": true,
  "allow_favorites": true,
  "face_search_enabled": true,
  "expires_at": "2027-06-18T23:59:59Z"
}
```

Response (201 Created):

```
{
  "id": "78b4092b-81d3-4f9e-a89e-3101b0ea5f12",
  "title": "Amalfi Coast Villa Cimbrone Editorial",
  "slug": "amalfi-coast-villa-cimbrone-editorial-9f201a",
  "client_name": "Elena & Matteo Rossi",
  "client_email": "elena.rossi@wedding.it",
  "event_date": "2026-06-18",
  "description": "Fine art sunset ceremony overlooking the Tyrrhenian Sea.",
  "template_id": "editorial",
  "status": "active",
  "visibility": "public",
  "is_password_protected": true,
  "allow_downloads": true,
  "allow_favorites": true,
  "face_search_enabled": true,
  "share_token": "uY48s_902kLSA...32chars",
  "share_url": "http://api.domain.com/api/storage/shared-galleries/uY48s_902kLSA.../",
  "public_url": "http://api.domain.com/api/storage/public/galleries/amalfi-coast-villa-cimbrone-editorial-9f201a/",
  "views_count": 0,
  "downloads_count": 0,
  "favorites_count": 0,
  "media_count": 0,
  "created_at": "2026-09-12T07:20:00Z"
}
```

5.2 List Galleries (Photographer Studio Dashboard)**Method:** GET**Endpoint:** /api/storage/galleries/**Auth:** Bearer Token (Photographer)**Query Parameters:**

- search: Search in title or client_name (e.g. `?search=Amalfi`)
- status: Filter by status (`active`, `draft`, `delivered`, `archived`)
- template_id: Filter by template (`editorial`, `masonry`, etc.)

Response (200 OK):

```
[
  {
    "id": "78b4092b-81d3-4f9e-a89e-3101b0ea5f12",
    "title": "Amalfi Coast Villa Cimbrone Editorial",
    "slug": "amalfi-coast-villa-cimbrone-editorial-9f201a",
    "client_name": "Elena & Matteo Rossi",
    "template_id": "editorial",
    "status": "active",
    "media_count": 48,
    "views_count": 142,
    "favorites_count": 28,
    "cover_image_url": "http://api.domain.com/media/galleries/cover.jpg"
  }
]
```

5.3 Get Gallery Detail & Media Items**Method:** GET

Endpoint: `/api/storage/galleries/<uuid:id>/`**Auth:** Bearer Token (Owner)**Response (200 OK):**

```
{
  "id": "78b4092b-81d3-4f9e-a89e-3101b0ea5f12",
  "title": "Amalfi Coast Villa Cimbrone Editorial",
  "slug": "amalfi-coast-villa-cimbrone-editorial-9f201a",
  "template_id": "editorial",
  "client_name": "Elena & Matteo Rossi",
  "client_email": "elena.rossi@wedding.it",
  "views_count": 142,
  "downloads_count": 19,
  "favorites_count": 28,
  "media_items": [
    {
      "id": "119e8400-e29b-41d4-a716-446655440001",
      "media_type": "photo",
      "title": "Sunset Spread Over Belvedere",
      "original_filename": "CIMBRONE_0194.jpg",
      "thumbnail_url": "http://api.domain.com/media/galleries/.../thumbnails/119e8400.
jpg",
      "preview_url": "http://api.domain.com/media/galleries/.../previews/119e8400.jpg"
    },
    {
      "download_url": "/api/storage/galleries/media/119e8400.../download/",
      "width": 6000,
      "height": 4000,
      "aspect_ratio": 1.5,
      "display_order": 1,
      "is_cover": true,
      "is_favorite": true,
      "processing_status": "ready"
    }
  ],
  {
    "id": "229e8400-e29b-41d4-a716-446655440002",
    "media_type": "video",
    "title": "4K Teaser Highlight",
    "original_filename": "Teaser_Highlight_4K.mp4",
    "duration": "02:45",
    "video_embed_url": "https://player.vimeo.com/video/987654321",
    "thumbnail_url": "http://api.domain.com/media/galleries/.../thumbnails/229e8400.
jpg",
    "preview_url": "http://api.domain.com/media/galleries/.../previews/229e8400.jpg"
  },
  {
    "download_url": "/api/storage/galleries/media/229e8400.../download/",
    "width": 3840,
    "height": 2160,
    "aspect_ratio": 1.778,
    "display_order": 2,
    "is_cover": false,
    "is_favorite": false,
    "processing_status": "ready"
  }
  }
}
```

5.4 Update Gallery Settings

Method: **PUT or PATCH****Endpoint:** `/api/storage/galleries/<uuid:id>/`**Auth:** Bearer Token (Owner)**Request Body:** Any subset of gallery fields (e.g. ``allow_downloads``, ``status``, ``title``).

5.5 Switch Layout Template ('editorial' | 'masonry' | 'cinematic' | 'minimal')

Method: **PATCH****Endpoint:** `/api/storage/galleries/<uuid:gallery_id>/template/`**Auth:** Bearer Token (Owner)**Request Body (JSON):**

```
{
  "template_id": "cinematic"
}
```

Response (200 OK):

```
{
  "status": "success",
  "template_id": "cinematic",
  "message": "Gallery layout template updated to cinematic."
}
```

5.6 Assign Gallery Cover Media**Method:** POST**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/set-cover/**Auth:** Bearer Token (Owner)**Request Body (JSON):**

```
{
  "media_id": "119e8400-e29b-41d4-a716-446655440001"
}
```

Response (200 OK):

```
{
  "status": "success",
  "cover_media_id": "119e8400-e29b-41d4-a716-446655440001",
  "message": "Cover media updated successfully."
}
```

5.7 Reorder Media Items (Sequence Ordering)**Method:** POST**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/reorder-media/**Auth:** Bearer Token (Owner)**Request Body (JSON):**

```
{
  "orders": [
    { "media_id": "119e8400-e29b-41d4-a716-446655440001", "display_order": 1 },
    { "media_id": "229e8400-e29b-41d4-a716-446655440002", "display_order": 2 }
  ]
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Reordered 2 media items."
}
```

5.8 Regenerate / Revoke Cryptographic Share Token

Generate New Token:

POST /api/storage/galleries/<uuid:gallery_id>/share/

Body: { "expires_in_days": 60 }

Response: { "share_token": "new_random_token...", "share_url": "..." }

Revoke Token:

DELETE /api/storage/galleries/<uuid:gallery_id>/share/

Response: { "status": "success", "message": "Share token revoked." }**6. DIRECT-TO-CLOUD STORAGE & UPLOAD PIPELINE****6.1 Direct Upload Initialization (Supports Batch Stills & Videos)****Method:** POST**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/upload-init/**Auth:** Bearer Token (Owner)**Rate Limit:** 60 / minute

Request Body (JSON):

```
{
  "files": [
    {
      "filename": "CEREMONY_001.JPG",
      "file_size": 24500000,
      "mime_type": "image/jpeg",
      "media_type": "photo"
    },
    {
      "filename": "HIGHLIGHT_REEL.MP4",
      "file_size": 180000000,
      "mime_type": "video/mp4",
      "media_type": "video"
    }
  ]
}
```

Response (200 OK):

```
{
  "files": [
    {
      "media_id": "550e8400-e29b-41d4-a716-446655440000",
      "reservation_id": "89b91e92-e7f0-4fa8-a1b7-d1e9f1938910",
      "storage_key": "galleries/78b4092b.../originals/550e8400.jpg",
      "upload_url": "https://bucket.s3.amazonaws.com/galleries/...&X-Amz-Signature=...",
      "http_method": "PUT",
      "headers": { "Content-Type": "image/jpeg" },
      "expires_in": 7200
    },
    {
      "media_id": "660e8400-e29b-41d4-a716-446655440000",
      "reservation_id": "99b91e92-e7f0-4fa8-a1b7-d1e9f1938910",
      "storage_key": "galleries/78b4092b.../originals/660e8400.mp4",
      "upload_url": "https://bucket.s3.amazonaws.com/galleries/...&X-Amz-Signature=...",
      "http_method": "PUT",
      "headers": { "Content-Type": "video/mp4" },
      "expires_in": 7200
    }
  ]
}
```

6.2 Direct Upload Confirmation & Worker Handoff**Method:** **POST****Endpoint:** `/api/storage/galleries/<uuid:gallery_id>/upload-confirm/`**Auth:** Bearer Token (Owner)**Request Body (JSON):**

```
{
  "media_id": "550e8400-e29b-41d4-a716-446655440000",
  "reservation_id": "89b91e92-e7f0-4fa8-a1b7-d1e9f1938910",
  "actual_file_size": 24500000,
  "width": 6000,
  "height": 4000,
  "aspect_ratio": 1.5,
  "duration": "00:00",
  "video_embed_url": null
}
```

Response (201 Created):

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "status": "processing",
  "message": "Media upload confirmed and indexed for Celery derivative processing."
}
```

6.3 Standard Multipart Upload (Fallback for Direct HTTP Form Uploads)**Method:** **POST****Endpoint:** `/api/storage/galleries/<uuid:gallery_id>/upload/`**Auth:** Bearer Token (Owner)

Content-Type: multipart/form-data

Body:

```
files: [file1.jpg, file2.jpg, ...]
media_type: "photo" (optional, default "photo")
```

6.4 Media Item Detail & Deletion

- GET /api/storage/galleries/media/<uuid:media_id>/ (Get item metadata)
- DELETE /api/storage/galleries/media/<uuid:media_id>/ (Permanently delete item)

Response (204 No Content)

6.5 Bulk Delete Media Items

Method: POST

Endpoint: /api/storage/galleries/media/bulk-delete/

Auth: Bearer Token (Owner)

Request Body (JSON):

```
{
  "media_ids": [
    "550e8400-e29b-41d4-a716-446655440000",
    "660e8400-e29b-41d4-a716-446655440000"
  ]
}
```

Response (200 OK):

```
{
  "status": "success",
  "deleted_count": 2,
  "message": "Successfully deleted 2 media items."
}
```

6.6 Toggle Studio Favorite / Star Flag

Method: POST

Endpoint: /api/storage/galleries/media/<uuid:media_id>/favorite/

Auth: Bearer Token (Owner)

Response (200 OK):

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "is_favorite": true,
  "message": "Media item starred as favorite."
}
```

7. CLIENT PUBLIC ACCESS, PIN SECURITY & PROOFING WORKFLOW

7.1 Public / Guest Gallery View by Slug or ID

Method: GET

Endpoint: /api/storage/public/galleries/<str:slug_or_id>/

Auth: None (Public)

Headers:

X-Gallery-PIN: "1234" (if gallery has PIN protection)

Response (200 OK):

```
{
  "id": "78b4092b-81d3-4f9e-a89e-3101b0ea5f12",
  "title": "Amalfi Coast Villa Cimbrone Editorial",
  "slug": "amalfi-coast-villa-cimbrone-editorial-9f201a",
  "template_id": "editorial",
  "client_name": "Elena & Matteo Rossi",
  "event_date": "2026-06-18",
  "photographer_studio": "LUMIÈRE STUDIO & ATELIER",
  "photographer_name": "Sarang Varma",
  "is_password_protected": false,
  "allow_downloads": true,
  "allow_favorites": true,
  "face_search_enabled": true,
  "media": [
    {
      "id": "119e8400-e29b-41d4-a716-446655440001",
      "media_type": "photo",
      "thumbnail_url": "http://api.domain.com/media/.../119e8400.jpg",
      "preview_url": "http://api.domain.com/media/.../119e8400.jpg",
      "download_url": "/api/storage/galleries/media/119e8400.../download/",
      "width": 6000,
      "height": 4000,
      "aspect_ratio": 1.5,
      "display_order": 1,
      "is_favorite": false
    }
  ]
}
```

7.2 Verify Gallery PIN Code**Method:** POST**Endpoint:** /api/storage/public/galleries/<str:slug_or_id>/verify-pin/**Auth:** None (Public)**Rate Limit:** 10 / minute (Brute force protection)**Request Body (JSON):**

```
{
  "pin": "Cimbrone2026!"
}
```

Response (200 OK):

```
{
  "status": "success",
  "valid": true,
  "message": "PIN verified successfully."
}
```

7.3 Access via Cryptographic Share Token**Method:** GET**Endpoint:** /api/storage/shared-galleries/<str:share_token>/**Auth:** None (Valid token acts as credential)**7.4 Verify Token Password****Method:** POST**Endpoint:** /api/storage/shared-galleries/<str:share_token>/verify-password/**Body:** { "password": "SecretPassword" }**7.5 Save Draft Client Selections / Favorites****Method:** POST**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/client-selections/**Auth:** None (Client session) or Authenticated Client

Request Body (JSON):

```
{
  "client_email": "elena.rossi@wedding.it",
  "client_name": "Elena Rossi",
  "selected_media_ids": [
    "119e8400-e29b-41d4-a716-446655440001",
    "229e8400-e29b-41d4-a716-446655440002"
  ],
  "client_notes": "First pass favorites for album spread."
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Client selections updated.",
  "selection": {
    "id": "cc9e8400-e29b-41d4-a716-446655440088",
    "client_email": "elena.rossi@wedding.it",
    "client_name": "Elena Rossi",
    "status": "draft",
    "selected_count": 2,
    "client_notes": "First pass favorites for album spread."
  }
}
```

7.6 List Gallery Client Selections (Photographer Studio View)

Method: GET**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/client-selections/**Auth:** Bearer Token (Owner)**Response (200 OK):**

```
[
  {
    "id": "cc9e8400-e29b-41d4-a716-446655440088",
    "client_email": "elena.rossi@wedding.it",
    "client_name": "Elena Rossi",
    "status": "submitted",
    "selected_count": 2,
    "client_notes": "Please retouch veil in photo #1.",
    "submitted_at": "2026-09-12T10:14:00Z"
  }
]
```

7.7 Submit Proofing Selections to Studio

Method: POST**Endpoint:** /api/storage/galleries/<uuid:gallery_id>/client-selections/<uuid:selection_id>/submit/**Auth:** None (Client session)**Request Body (JSON):**

```
{
  "client_notes": "Approved final 45 images for print. Retouch veil in #1."
}
```

Response (200 OK):

```
{
  "status": "success",
  "message": "Proofing selection submitted to photographer studio successfully.",
  "selection": {
    "id": "cc9e8400-e29b-41d4-a716-446655440088",
    "status": "submitted",
    "submitted_at": "2026-09-12T10:14:00Z"
  }
}
```

8. DOWNLOAD PIPELINE & ASYNCHRONOUS BULK ZIP

8.1 Single Media Signed Download URL

Method: GET**Endpoint:** /api/storage/galleries/media/<uuid:media_id>/download/

Auth: None (Public if gallery open) or Bearer Token

Rate Limit: 120 / minute

Query Parameters:

- share_token: "<token>" (if accessing through token link)

Response (200 OK):

```
{
  "download_url": "https://cdn.domain.com/galleries/.../original.jpg?X-Amz-Signature=.",
  "filename": "CIMBRONE_0194.jpg",
  "file_size": 24500000,
  "mime_type": "image/jpeg",
  "expires_in": 900
}
```

8.2 Initiate Master or Selective Bulk ZIP Job

Method: **POST**

Endpoint: `/api/storage/galleries/<uuid:gallery_id>/bulk-download/`

Auth: None (with share_token) or Authenticated User

Rate Limit: 5 / minute

Request Body (JSON):

```
{
  "download_type": "selected_subset", // "master_all" or "selected_subset"
  "media_ids": [
    "119e8400-e29b-41d4-a716-446655440001",
    "229e8400-e29b-41d4-a716-446655440002"
  ],
  "share_token": "<token>" // Optional if gallery is protected
}
```

Response (202 Accepted):

```
{
  "job_id": "fa9e8400-e29b-41d4-a716-446655440077",
  "status": "pending",
  "download_type": "selected_subset",
  "selected_count": 2,
  "poll_url": "/api/storage/galleries/bulk-download-jobs/fa9e8400-e29b-41d4-a716-446655440077/",
  "message": "Bulk ZIP compilation queued. Poll the status URL for completion."
}
```

8.3 Poll Bulk Download Job Status

Method: **GET**

Endpoint: `/api/storage/galleries/bulk-download-jobs/<uuid:job_id>/`

Auth: None (or matching requester)

Response (When in progress - 200 OK):

```
{
  "id": "fa9e8400-e29b-41d4-a716-446655440077",
  "status": "processing",
  "progress_percent": 65,
  "download_url": null
}
```

Response (When ready - 200 OK):

```
{
  "id": "fa9e8400-e29b-41d4-a716-446655440077",
  "status": "ready",
  "progress_percent": 100,
  "file_size": 48200000,
  "download_url": "https://cdn.domain.com/archives/Amalfi_Selections.zip?sig=...",
  "expires_at": "2026-09-13T10:00:00Z"
}
```

9. AI BIOMETRIC FACE RECOGNITION & DISCOVERY

9.1 Face Selfie Search (Gallery-Isolated Discovery)

Method: **POST****Endpoint:** `/api/storage/galleries/<uuid:gallery_id>/face-search/`**Auth:** None (Public with share_token) or Authenticated Client**Rate Limit:** 15 / minute

Content-Type: multipart/form-data

Body:

```
selfie: <image binary of guest/client selfie (JPEG/PNG)>
share_token: "<token>" (if gallery requires access token)
```

Response (200 OK):

```
{
  "total_matches": 14,
  "matches": [
    {
      "id": "119e8400-e29b-41d4-a716-446655440001",
      "thumbnail_url": "http://api.domain.com/media/.../119e8400.jpg",
      "preview_url": "http://api.domain.com/media/.../119e8400.jpg",
      "download_url": "/api/storage/galleries/media/119e8400.../download/",
      "confidence": 0.94,
      "width": 6000,
      "height": 4000
    }
  ]
}
```

10. STORAGE ACCOUNT METRICS & QUOTA AUDIT

10.1 Detailed Storage Breakdown & Quota Statistics

Method: **GET****Endpoint:** `/api/storage/account/storage/` (or `/api/account/storage/`)**Auth:** Bearer Token (Photographer)**Response (200 OK):**

```
{
  "storage_limit_bytes": 128849018880,    // 120 GB
  "storage_used_bytes": 12884901888,      // ~12 GB
  "storage_reserved_bytes": 1500000,      // 1.5 MB in-flight
  "storage_remaining_bytes": 115962616992, // ~108 GB remaining
  "usage_percentage": 10.0,
  "total_galleries": 6,
  "max_galleries": 0,                     // Unlimited on Pro plan
  "plan_name": "Pro Studio Annual",
  "face_search_enabled": true
}
```

10.2 Storage Audit & Reconciliation CLI (Django Management Command)

Command:

```
python manage.py reconcile_storage_usage
python manage.py reconcile_storage_usage --photographer-id 1
```

11. EVENT PHOTO SHARING & GUEST QR ACCESS (`/api/storage/events/` & `/api/storage/share/`)

Quick guest sharing workflows for high-volume wedding receptions and events.

11.1 Create Shared Event

Method: **POST****Endpoint:** `/api/storage/events/` (or `/api/events/`)**Auth:** Bearer Token (Photographer)

Request Body (JSON):

```
{
  "title": "Villa d'Este Welcome Cocktail",
  "event_type": "Wedding",
  "event_date": "2026-07-22",
  "location": "Cernobbio, Lake Como",
  "is_password_protected": true,
  "pin_code": "8899",
  "allow_bulk_download": true
}
```

Response (201 Created):

```
{
  "id": 1,
  "title": "Villa d'Este Welcome Cocktail",
  "access_code": "VDE-9921",
  "shareable_url": "http://domain.com/share/VDE-9921/",
  "qr_code_url": "/api/storage/share/VDE-9921/qr/",
  "photo_count": 0
}
```

11.2 List Shared Events

Method: GET**Endpoint:** /api/storage/events/**Auth:** Bearer Token (Photographer)

11.3 Bulk Photo Upload to Event (Multipart)

Method: POST**Endpoint:** /api/storage/events/<int:event_id>/upload/**Auth:** Bearer Token

Content-Type: multipart/form-data

Body:

```
photos: [photo1.jpg, photo2.jpg, ...]
caption: "Reception" (optional)
```

11.4 Bulk Photo ZIP Import & Extraction

Method: POST**Endpoint:** /api/storage/events/<int:event_id>/upload-zip/**Auth:** Bearer Token

Content-Type: multipart/form-data

Body:

```
zip_file: <archive.zip>
```

Response (201 Created):

```
{
  "message": "ZIP extracted. 85 photos imported successfully.",
  "imported_count": 85
}
```

11.5 Bulk Delete Event Photos

Method: POST**Endpoint:** /api/storage/events/<int:event_id>/photos/bulk-delete/**Auth:** Bearer Token

Body (JSON):

```
{ "photo_ids": [101, 102, 103] }
```

11.6 Public Guest View by Access Code

Method: GET**Endpoint:** /api/storage/share/<str:access_code>/**Auth:** None (Public)

Headers:

X-Event-PIN: "8899" (if event has PIN)

Response (200 OK):

```
{
  "id": 1,
  "title": "Villa d'Este Welcome Cocktail",
  "access_code": "VDE-9921",
  "photo_count": 85,
  "allow_bulk_download": true,
  "photos": [
    {
      "id": 201,
      "image": "http://api.domain.com/media/events/photo201.jpg",
      "thumbnail": "http://api.domain.com/media/events/thumbs/photo201.jpg",
      "download_url": "/api/storage/photos/201/download/"
    }
  ]
}
```

11.7 Verify Event PIN Code

Method: **POST**
Endpoint: /api/storage/share/<str:access_code>/verify-pin/
Body (JSON): { "pin": "8899" }

11.8 Streamed Instant All Photos ZIP Download

Method: **GET**
Endpoint: /api/storage/share/<str:access_code>/download-all/
Auth: None (Public)
Response: Binary streaming archive `application/zip`

11.9 Single Event Photo Download

Method: **GET**
Endpoint: /api/storage/photos/<int:photo_id>/download/
Auth: None (Public)

11.10 Instant Scannable QR Code Image

Method: **GET**
Endpoint: /api/storage/share/<str:access_code>/qr/
Auth: None (Public)
Response: `image/png` binary barcode

12. RATE LIMITING, SECURITY PROTOCOLS & ERROR CODES

12.1 Rate Limit Matrix:

Throttle Scope Rate Limit Target Endpoints

anon	200 / minute	Public endpoint browsing & guest access
user	1000 / minute	Authenticated studio API interactions
face_search	15 / minute	Selfie biometric feature search
upload	60 / minute	Direct upload init & multipart uploads
download	120 / minute	Single media signed downloads
bulk_download	5 / minute	Async bulk ZIP compilation requests

pin_verify 10 / minute Public gallery and event PIN checks

12.2 Standard Error Response Payload:
When any API call encounters an error, a structured JSON payload is returned:

```
{
  "code": "ERROR_CODE_STRING",
  "detail": "Human-readable explanation of why the request failed."
}
```

```
}
```

Common Error Codes:

- 400 VALIDATION_ERROR: Missing required fields or malformed payload.
- 401 AUTHENTICATION_REQUIRED: Missing or expired JWT Bearer token.
- 403 GALLERY_ACCESS_DENIED: Gallery is private or password required.
- 403 DOWNLOAD_DISABLED: Studio has disabled downloads for this gallery.
- 403 FACE_SEARCH_DISABLED: Biometric face search is toggled off.
- 404 NOT_FOUND: Gallery, media item, or job ID does not exist.
- 410 LINK_EXPIRED: The requested cryptographic share token has expired.
- 413 STORAGE_LIMIT_EXCEEDED: Requested upload exceeds subscription storage limit.
- 429 RATE_LIMIT_EXCEEDED: Scoped throttle threshold exceeded; retry after backoff.

END OF API SPECIFICATION (VERSION 2.0)